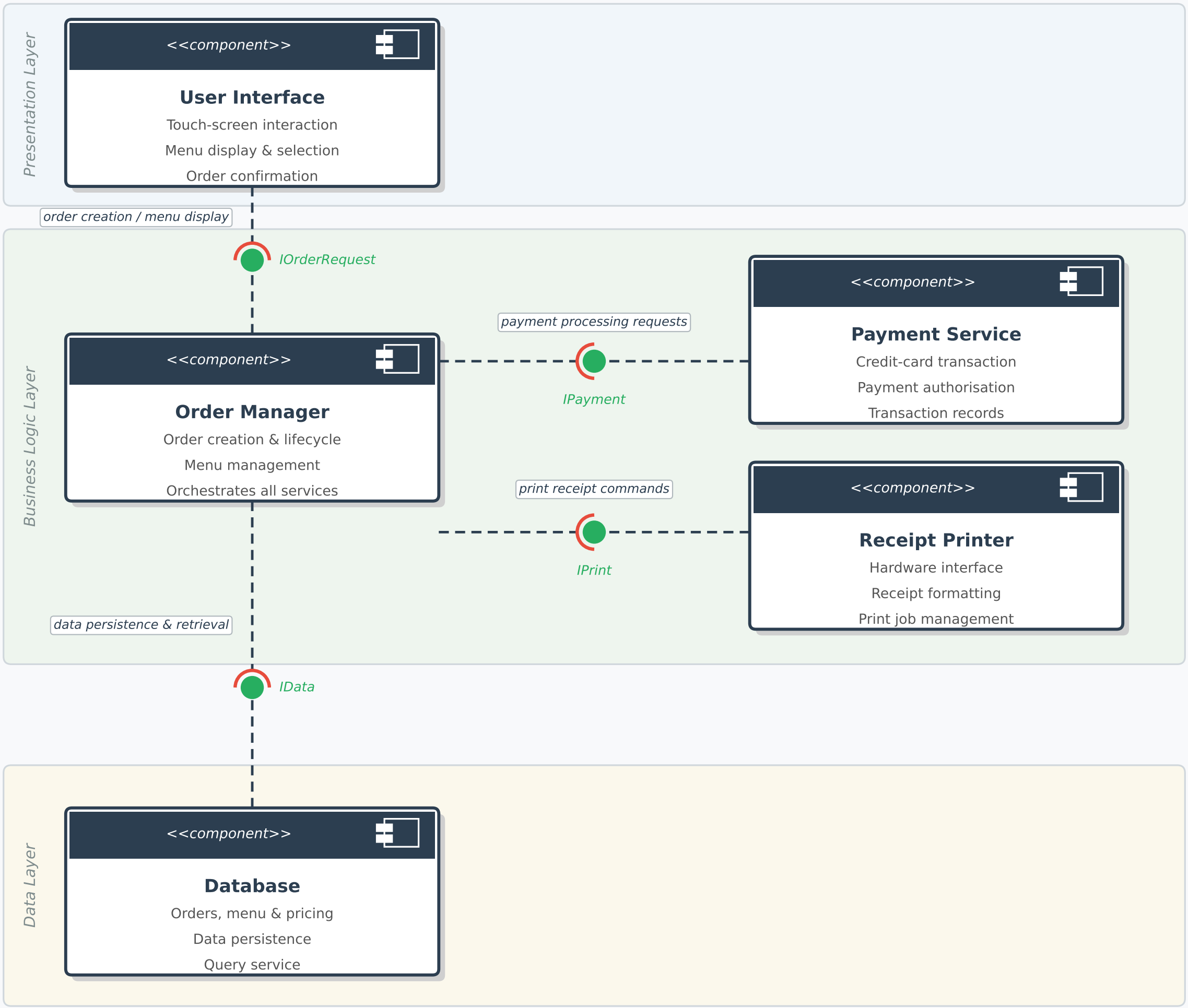
UML Component Diagram — Self-Service Coffee Kiosk System

PES1UG24CS086 | Layered Architecture



Lab 3: Architectural Justification — Self-Service Coffee Kiosk

PES1UG24CS086 · Internal Microservice Catalog & Health Portal · DevOps Engineer / System Architect

1. Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

2. Architectural Style Comparison

Architecture	Pros	Cons	Fit?
Layered	Clear separation; simple; testable	Layer overhead; tight coupling	✔ Best Fit
Microservices	Independent scaling; fault isolation	Network latency; operationally complex; overkill	✘ Overkill
Client-Server	Centralised control; simple deployment	Single point of failure; scalability bottleneck	~ Partial

3. Two Scenario-Specific Reasons

Reason 1 — Well-scoped, single-unit system. The kiosk is a standalone device with a tightly defined workflow: menu selection → order creation → payment → receipt printing. Layered architecture maps directly onto this linear flow (Presentation → Business Logic → Data), keeping the codebase simple to develop and test without the operational overhead of distributed services.

Reason 2 — Strict 60-second order-completion requirement. All layers execute within the same process, so inter-layer calls are in-process function invocations rather than network round-trips. This eliminates serialisation and network latency, ensuring the full order-to-receipt cycle fits the 60-second SLA.

4. Security Advantage

The **Payment Service sits exclusively in the Business Logic layer** and is never directly reachable from the Presentation layer. The UI communicates only with the Order Manager, so credit-card data never passes through presentation-layer code, reducing the PCI-DSS attack surface. Layer boundaries serve as natural enforcement points where input validation and sanitisation are applied before any sensitive payment operation is invoked.

5. Performance Benefit

All five components communicate through **in-process interface calls**, keeping end-to-end latency in the low-millisecond range. The Database component uses a lightweight embedded store (e.g., SQLite) co-located on the kiosk, so menu and pricing lookups are effectively memory-speed reads, comfortably supporting the under-60-second order SLA on modest kiosk hardware.